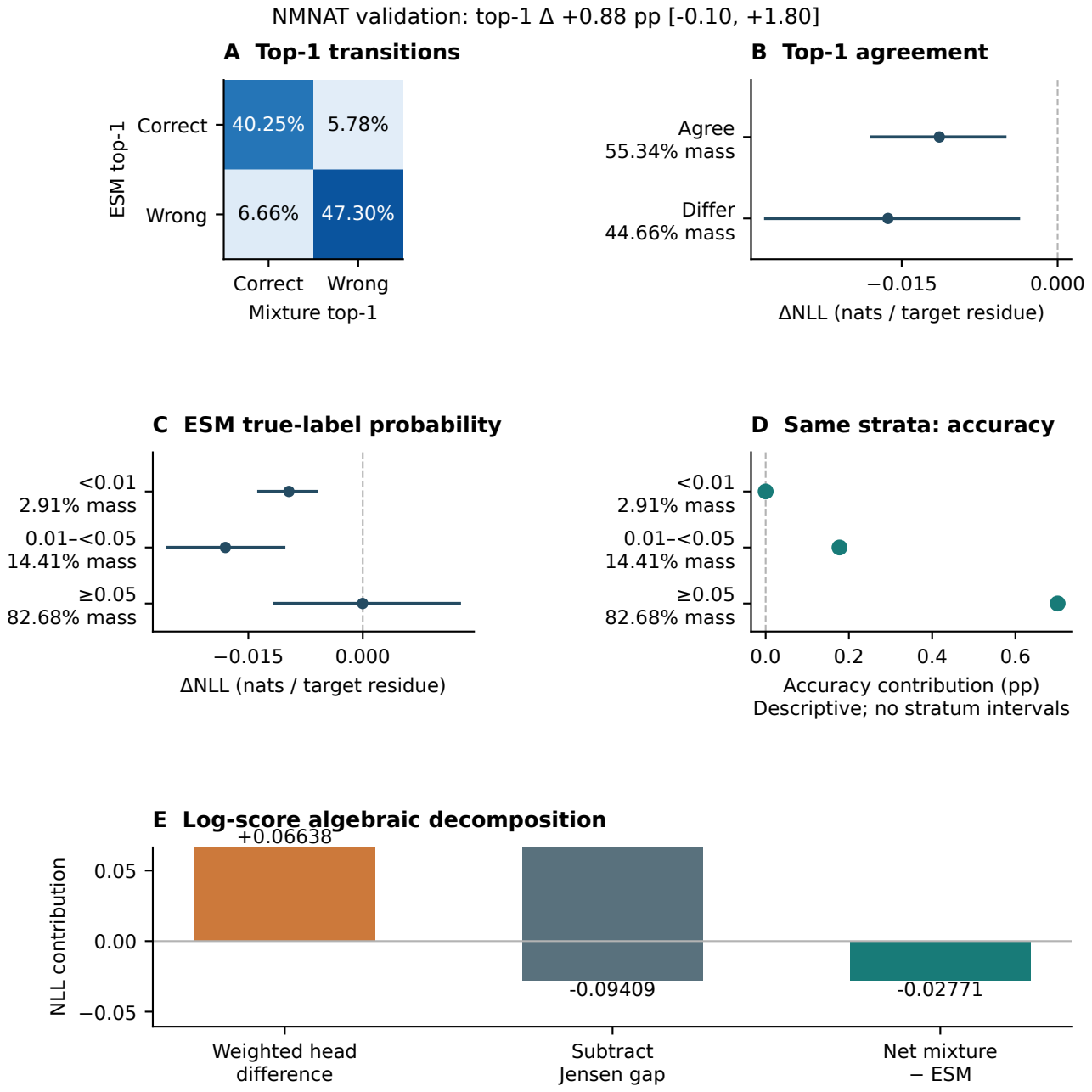
NMNAT validation: top-1 Δ +0.88 pp [-0.10, +1.80]
A Top-1 transitions
ESM top-1
Correct
Wrong
40.25%
5.78%
6.66%
47.30%
Correct
Wrong
Mixture top-1
B Top-1 agreement
Agree
55.34% mass
Differ
44.66% mass
−0.015
0.000
ΔNLL (nats / target residue)
C ESM true-label probability
<0.01
2.91% mass
0.01–<0.05
14.41% mass
≥0.05
82.68% mass
−0.015
0.000
ΔNLL (nats / target residue)
D Same strata: accuracy
<0.01
2.91% mass
0.01–<0.05
14.41% mass
≥0.05
82.68% mass
0.0
0.2
0.4
0.6
Accuracy contribution (pp)
Descriptive; no stratum intervals
E Log-score algebraic decomposition
NLL contribution
0.05
0.00
−0.05
+0.06638
-0.09409
-0.02771
Weighted head difference
Subtract Jensen gap
Net mixture − ESM